

JUNTANG WANG

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EDUCATION

Harvard University

M.S. in Data Science

2026 – 2027

Cambridge, MA

Duke University & Duke Kunshan University (Dual Degree)

B.S. in Interdisciplinary Studies; B.S. in Applied Math & Computational Science (CS Track)

Durham, NC; Kunshan, China

2022 – 2026

- GPA: 3.8/4.0 (Major: 4.0/4.0); A⁺ Courses: Deep Learning, Machine Learning, Matrix/Graph/Network Analysis, Databases.

PUBLICATIONS & OPEN-SOURCE SOFTWARE

[†]Equal contribution. 6 additional manuscripts under review (1× ICML, 3× KDD, 1× ECCV, 1× PNAS Nexus).

Juntang Wang[†], Hao Wu[†], Yihan Wang[†], Dongmian Zou, and Shixin Xu. (2026). “Mixing Configurations for Downstream Prediction.” *ICML 2026*. [arXiv:2510.19248](https://arxiv.org/abs/2510.19248)

Juntang Wang[†], Yihan Wang[†], Hao Wu[†], Dongmian Zou, and Shixin Xu. (2025). “Brain-Inspired Perspective on Configurations: Unsupervised Similarity and Early Cognition.” *BICS (Oral)*. [arXiv:2510.19229](https://arxiv.org/abs/2510.19229)

Shu Kit Eric Tam, Juntang Wang, Jianke Cen, Aleksandra Stryjska, Pascal Grange, and Sze Chai Kwok. (2025). “Can the mammalian circadian system adapt to the Martian photoperiod?” *18th Annual Meeting of the Chinese Neuroscience Society*.

Yihan Wang[†], Juntang Wang[†], Xinze Xu, Yihen Han, Qinyi Chen, Ghulam Hussain, and Xiawa Wang. (2025). “Analyzing temperature-induced phase transitions in $\text{Pb}_{10-x}\text{Cu}_x(\text{PO}_4)_6\text{O}$.” *17th International Conference on Materials Chemistry*.

mheatmap | [\[GitHub\]](https://github.com/jw853/mheatmap) | **680+** Stars | Sole developer: a Python pkg. for proportional heatmap visualization and spectral reordering.

RESEARCH EXPERIENCE

ML Research: Graph Methods, Generative Models, and Biomedical AI

Mar 2025 – Mar 2026

Signature Work; Research Assistant; PI: Prof. Shixin Xu, Prof. Dongmian Zou

Kunshan, China

- Produced 7 first/co-first-author papers (1× ICML, 1× BICS Oral; 1× ICML, 3× KDD, 1× ECCV under review) across biomedical AI, graph-based ML, neural rendering, and network security.
- Developed MixConfig (configuration-based feature augmentation) and PPIDM-II (physics-informed diffusion with abstention); proposed Variability-Gated Encoder improving joint accuracy 4.7% over Transformer on 65K-strain 16S rRNA benchmark.

EEG/EMG Time-Series Modeling for Neuroscience

Mar 2025 – Aug 2025

Summer Research Scholar; PI: Prof. Shu Kit Eric Tam

Kunshan, China

- Built a CNN achieving 90%+ accuracy on mouse EEG/EMG vigilance-state data, benchmarked against 10+ methods.
- Co-authored one SCI-journal manuscript (under review) and presented findings at a national neuroscience conference.

Graph and Spectral Methods for Hyperspectral Image Analysis

Jun 2024 – Dec 2024

Summer Research; Independent Study; PI: Prof. Xiaobai Sun

Durham, NC

- Benchmarked 5+ community-detection and clustering algorithms across 10+ hyperspectral imaging datasets.
- Released **mheatmap** and **sgtsnepi**, a pure-Python sklearn-compatible SG-t-SNE graph embedding implementation.

WORK EXPERIENCE

Data Scientist, Intern

Mar 2026 – May 2026

Siemens Healthineers

Shanghai, China

- Applied an InceptionTime 1-D CNN for CT X-ray tube fault diagnosis from kV/mA signals; 97% accuracy on 5 fault classes.
- Built a BERT-NER + Bedrock LLM pipeline on 200+ service docs: 99.4% PII recall, 89.4% CT-component accuracy.

Credit Analyst, Intern

Jun 2023 – Aug 2023

Huaxia Bank

Kunshan, China

- Assessed creditworthiness of 50+ local electronics firms, synthesizing findings to support loan risk decisions.
- Compiled 50+ audit reports with financial statements and industry benchmarks, streamlining the loan assessment workflow.

ADDITIONAL INFORMATION

Honors: Duke Graduation with Distinction; Kaggle Bronze (Stanford RNA 3D Folding); Dean’s List with Distinction (×3)

Service: Resident Assistant at DKU, Orientation Peer at Duke, Kendo Club Training Leader

Skills: Python, SQL, MATLAB, Julia, Java, C; scikit-learn, PyTorch, NumPy, pandas; PostgreSQL, MongoDB; Git, $\text{\LaTeX}$

Teaching: TA for CS 521: Matrix/Graph/Network Analysis (**Graduate**), MATH 302 Numerical Analysis, MATH 101 Calculus